MSc Finance 20127

Corporate Finance (Business Valuation)
Growth & Value

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Estimating Growth

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Estimating Growth: Definition and Approaches I

Assets		Liabilities	
Assets in place	Existing Investments Generate Cash Flows Today	(Net) Debt	Borrowed Money
Growth Opportunities	Investments to be made Generate Cash Flows Tomorrow	Equity	Owners' Funds

- Value of the Firm (either Equity or Asset Side) using DCF approach

$$\text{Value}_0 = \sum_{t=1}^{n} \frac{\mathbb{E}\left[\text{CF}_t\right]}{(1+k_c)^t} + \frac{\text{Terminal Value}_n}{(1+k_c)^n}$$

where:

$\mathbb{E}\left[\text{CF}_t\right]$: Expected Cash Flow to Equity in year t (either Equity or Asset side)

k_c: Cost of capital (either Equity or Asset side)

TV: Terminal Value at the end of the explicit estimate of CF

- Under stable growth, $\text{TV}_n = \frac{\mathbb{E}[\text{CF}_{n+1}]}{k_c - g}$, where g is the estimate of long run CF growth

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Estimating Growth: Definition and Approaches II

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- There are three main approaches to estimate growth g
  1. **Historical**: g is estimated based on the historical growth in Earnings per Share (EPS)
  2. **Analysts' Estimates**: g is based on analysts' estimates of EPS growth
  3. **Fundamental**: g is estimated looking at fundamentals of EPS's growth
     - ✓ Reinvestments (non-cash working capital and net capex)
     - ✓ Return on investments (what returns such projects are making)

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Estimating Growth: Historical Approach

- Historical Growth in EPS can be estimated in different ways
  1. **Arithmetic Average**: $g_t = \frac{1}{n}\sum_{i=1}^{n} g_{t-i} = \frac{1}{n}\sum_{i=1}^{n} \frac{[\text{NI}_{(t-i)} - \text{NI}_{(t-i-1)}]}{\text{NI}_{(t-i-1)}}$
  2. **Geometric Average**: $g_t = \left[\prod_{i=1}^{n}(1+g_{t-i})\right]^{\frac{1}{n}} - 1 = \left(\frac{\text{NI}_t}{\text{NI}_{(t-n)}}\right)^{\frac{1}{n}} - 1$
  3. **Log-linear OLS Regression Models**: $\ln(\text{EPS}) = a + b \cdot t$, where t is time and $\widehat{b}$ yields the percentage change in EPS per unit of time
  4. **Time Series Models**: (Box and Jenkins 1976; Foster 1977; Griffin 1977)

- Historical rates can be sensitive to
  1. Estimation period; needs to be chosen carefully, especially for young / high growth firms
  2. Negative earnings: if some EPS are negative, can't compute ln(EPS) for log-linear regression
  3. Earnings volatility: it may be high especially for young / high growth industries. Revenues' growth is more stable and serially correlated than earnings' growth, so it may be worth to use revenues' growth as a check
  4. Size effect: there is growth's convexity as growth reduces while size increases
  5. Changes in EPS not driven by earnings (e.g., stock splits)



Estimating Growth: Historical Approach — Motorola (data in millions)

Year	Revenues	% Change
X	$22,245	
X+1	$27,037	21.54%
X+2	$27,973	3.46%
X+3	$29,794	6.51%
X+4	$29,398	−1.33%
X+5	$30,931	5.21%
Arithmetic Average		7.08%
Geometric Average		6.82%
Standard Deviation		8.61%

Year	EBITDA	% Change
X	$4,151	
X+1	$4,850	16.84%
X+2	$4,268	−12.00%
X+3	$4,276	0.19%
X+4	$3,019	−29.40%
X+5	$5,398	78.80%
Arithmetic Average		10.89%
Geometric Average		5.39%
Standard Deviation		41.56%

Year	EBIT	% Change
X	$2,604	
X+1	$2,931	12.56%
X+2	$1,960	−33.13%
X+3	$1,947	−0.66%
X+4	$822	−57.78%
X+5	$3,216	291.24%
Arithmetic Average		42.45%
Geometric Average		4.31%
Standard Deviation		141.78%

Year	EPS	% Change
X	$0.88	
X+1	$0.98	11.36%
X+2	$0.65	−33.67%
X+3	$0.66	1.54%
X+4	$0.14	−78.79%
X+5	$0.45	221.43%
Arithmetic Average		24.37%
Geometric Average		−12.55%
Standard Deviation		115.69%

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Estimating Growth: Analysts' Estimates Approach

- Analysts' consensus growth estimates available from online providers (e.g., I/B/E/S)
- A significant proportion of analysts' time is spent forecasting EPS, based on
  1. **Historical Growth** — see above
  2. **Public Information Other Than Earnings** affecting EPS' g
  3. **Recent Firm Specific Information** updating historical g's estimate
  4. **Recent Market/Industry Information** affecting future growth
  5. **Information Revealed by Competitors and/or Private**
- Analysts' estimates can be sensitive to
  1. Number of analysts covering a firm, which increases with (i) market cap, (ii) institutional holdings, and (iii) trading volume
  2. Extent of disagreement among analysts
  3. Independence of analysts' estimates
  4. Analysts' quality

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Estimating Growth: Analysts' Pitfalls

Analysts' five "deadly sins"

(i) Tunnel vision. Too focused on the industry

(ii) "Lemmingitis" (aka, herding). Urge to revise when others do, in same direction and magnitude as others

(iii) Stockholm Syndrome. Identification with the managers

(iv) Factophobia. Relying on "stories" rather than facts

(v) Dr. Jekill/Mr. Hyde. Bring business to investment banks

Estimating Growth: Fundamentals' Approach — I

The fundamentals' approach estimates growth as an endogenous variable

1. **EPS Growth**

$$g = \text{Retention Ratio}(b) \times \text{ROE}$$

Proof

- Define: Growth in Net Income, $g = \frac{\text{NI}_t - \text{NI}_{t-1}}{\text{NI}_{t-1}}$; $\text{ROE}_{t-1} = \frac{\text{NI}_{t-1}}{\text{BV}_{t-2}}$, where BV is Equity Book Value
- Assuming no equity injections, $\text{NI}_t = \text{BV}_{t-1} \times \text{ROE}_t = (\text{BV}_{t-2} + \text{Retained Earnings}_{t-1}) \times \text{ROE}_t$
- Assuming constant ROE ($\text{ROE}_{t-1} = \text{ROE}_t$) and substituting, obtain

$$g = \frac{\text{Retained Earnings}_{t-1}}{\text{NI}_{t-1}} \times \text{ROE} = \text{Retention Ratio} \times \text{ROE} = b \times \text{ROE}. \textbf{ QED}$$

- Implicit assumptions
  - Constant ROE over time
  - No equity issuance allowed, therefore NI Growth = EPS growth
  - No other comprehensive income in the book value of equity

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Estimating Growth: Fundamentals' Approach — II

The fundamentals' approach estimates growth as an endogenous variable

2. Net Income (NI) Growth

$$g = b \times \text{ROE}$$

Proof

- Similar steps as before, but now b is Equity Reinvestment Rate, as follows

$$\text{Equity Reinvested}_t = (\text{CAPEX}_t - \text{D\&A}_t) + \Delta\text{WC}_t - (\text{New Debt Issued}_t - \text{Debt Repaid}_t)$$

- Equity Reinvestment Rate: $b_t = \frac{\text{Equity Reinvested}_t}{\text{NI}_t}$
- Expected Growth Rate

$$g_{t+1} = b_t \times \text{ROE}. \textbf{ QED}$$

- Adjust all figures (R&D, operating leases, etc.) consistent with adjustments made to cash flows

▶ Implicit assumptions

- Constant ROE over time
- Equity issuance is allowed, so NI Growth $\neq$ EPS growth
- No other comprehensive income in the book value of equity

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Estimating Growth: Fundamentals' Approach — ROE

The fundamentals' approach requires always an estimate of Return on Equity (ROE)

- Following Modigliani Miller (with taxes):

$$k_A = k_E \cdot \frac{E}{D+E} + k_D \cdot (1-\tau_C) \cdot \frac{D}{D+E} \iff k_E = k_A + [k_A - k_D \cdot (1-\tau_C)] \cdot \frac{D}{E}$$

- In accounting terms:

$$\text{ROE} = \text{ROC} + [\text{ROC} - i \cdot (1-\tau_C)] \cdot \frac{D}{E}$$

where

$$\text{ROC = Return on Capital} = \frac{\text{EBIT} \cdot (1-\tau_C)}{\text{BV of Debt} - \text{Cash} + \text{BV of Equity}} = \frac{\text{EBIT} \cdot (1-\tau_C)}{\text{BV of Net Debt} + \text{BV of Equity}}$$

$$\frac{D}{E} = \frac{\text{BV of Net Debt}}{\text{BV of Equity}}$$

$$i = \frac{\text{Interest Expenses on (Net) Debt}}{\text{BV of (Net) Debt}}$$

- Implications
  - ROE is a function of returns on investments (ROC) undertaken
  - ROE changes with leverage (and financial risk)
  - The higher the cost of debt, the lower the ROE
  - Marginal ROE is different from average ROE

$$\longrightarrow \text{ addition to expected } g = \frac{\text{ROE}_t - \text{ROE}_{t-1}}{\text{ROE}_{t-1}}$$

Estimating Growth: Fundamentals' Approach — EBIT I

We can apply the fundamentals' approach to estimate growth of operating income

1. **Stable Return on Capital (ROC) Scenario**

$$g = b \times \text{ROC}$$

- b = Reinvestment Rate
- Reinvestment Rate = $\frac{(\text{CAPEX}_t - \text{D\&A}_t) + \Delta \text{WC}_t}{[\text{EBIT}_t \cdot (1 - \tau_C)]}$
- Return on Capital (ROC) = $\frac{\text{EBIT}_t \cdot (1 - \tau_C)}{\text{BV of Equity} + \text{BV of Debt} - \text{Cash}}$

▶ Implicit assumptions

- Constant ROC and EBIT margins over time
- *b* from recent financial statements is forward looking
- Accounting ROC is a good measure of required returns on assets (to be checked against industry average)
- No other comprehensive income (OCI) in the book value of equity

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Estimating Growth: Fundamentals' Approach — EBIT II

We can apply the fundamentals' approach to estimate growth of operating income

2. **Positive and Changing Return on Capital (ROC) Scenario**

$$g = b \times \text{ROC} + \Delta\% \text{ ROC}$$

- Same derivation as before; Δ % ROC = $\frac{\text{ROC}_t - \text{ROC}_{t-1}}{\text{ROC}_{t-1}}$

3. **Negative Return on Capital (ROC)/Changing Margins Scenario**

(i) Estimate growth rates in Revenues over time
- Use historical revenue growth to get estimates of revenue growth in the near future
- Decrease the growth rate as the firm becomes larger (growth's convexity)
- keep track of absolute revenues to make sure the growth is feasible

(ii) Estimate expected operating margins every year
- Set a target margin that the firm will move towards
- Adjust the current margin towards the target margin
- Use target margin in perpetuity

(iii) Estimate the capital to be invested to generate expected revenue growth / margins
- Estimate Sales-to-Capital Ratio: Investment Needs=Revenue Growth/Sales-to-Capital Ratio

(iv) Check the implied ROC is consistent with sustainable long-run growth in industry/economy

Estimating Growth: Recap

- Fundamental equation for growth
  - Dividend approach: $g = \text{Retention Ratio}(b) \cdot \text{ROE} + \Delta\%\ \text{ROE}$
  - FCFE approach: $g = \text{Equity Reinvestment Rate} \cdot \text{ROE} + \Delta\%\ \text{ROE}$
  - FCFF approach: $g = \text{Reinvestment Rate} \cdot \text{ROC} + \Delta\%\ \text{ROC}$

- In stable growth, firms have to reinvest to deliver the growth rate that you have forecast (i.e. Δ %ROE and Δ %ROC goes to zero)
  - Dividend approach: $g = \text{Retention Ratio}(b) \cdot \text{ROE}$
  - FCFE approach: $g = \text{Equity Reinvestment Rate} \cdot \text{ROE}$
  - FCFF approach: $g = \text{Reinvestment Rate} \cdot \text{ROC}$

- In stable growth, reinvestment rate is a function of growth and return on capital
  - Dividend approach: $\text{Retention Ratio}(b) = g/\text{ROE}$
  - FCFE approach: $\text{Equity Reinvestment Rate} = g/\text{ROE}$
  - FCFF approach: $\text{Reinvestment Rate} = g/\text{ROC}$

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$\Longrightarrow$ **Need to estimate both g and ROE/ROC: assumptions' consistency is key**

Terminal Value

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Terminal Value: Explicit Projections I

- Any asset value (financial or real) is a function of:
  1. The **Cash Flows** (CF) it can generate
  2. The **time distributions** of the cash flows

- The time horizon of CF projections plays a key role in valuation; it should:
  - Show the CF dynamics linked to time periods of corporate financial restructuring
  - Show the CF dynamics linked to extraordinary investment cycles
  - Analyze growth phases due to competitive imbalances
  - Describe performance dynamics in industries characterized by cyclical trends

- Finally, when the project's duration is tied to the availability of irreplaceable resources, renewals, or concession contracts, its time horizon can correspond to:
  - (i) The **exhaustion** of these resources
  - (ii) The **maturity** of concession contracts

Terminal Value: Explicit Projections II

- According to international practice, the length of the explicit projection period varies
  - From 7 to 10 years in low-capital-intensive sectors
  - Up to 10-15 years in industries whose specific assets have long economic/technical lives (hydroelectric, petrochemical, ...)

- Therefore, analysts usually begin with a business plan that does not exceed 3 to 5 years: the plan can then be further extended as necessary

- The plan extension needs to fulfill a double requirement
  1. The entire length of the business cycle should be covered
  2. The portion of value obtained by discounting CF of explicit annual projections should be **reasonable** as compared to the discounted value of the Terminal Value
     - This share generally changes according to the business area under consideration

Terminal Value: Cash Flows Valuations I

- A firm potentially has an infinite life $\Longrightarrow$ Its value is the present value of CF forever

$$V_t = \sum_{s=t+1}^{n} \frac{\mathbb{E}_t\left[FCF_s\right]}{(1+k_C)^{s-t}}$$

for $n \longrightarrow \infty$

- Since we cannot estimate FCF forever, we estimate FCF explicitly for a "growth period" and then we estimate a Terminal Value (TV) to capture any subsequent FCF thereafter

$$V_t = \sum_{s=t+1}^{n} \frac{\mathbb{E}_t\left[FCF_s\right]}{(1+k_C)^{s-t}} + \frac{TV_n}{(1+k_C)^{n}}$$

where:

$\mathbb{E}_t\left[FCF_s\right]$: Expectation at time t of Free Cash Flow in period s (Equity or Asset Side)

k_C: Cost of Capital (Equity or Asset Side)

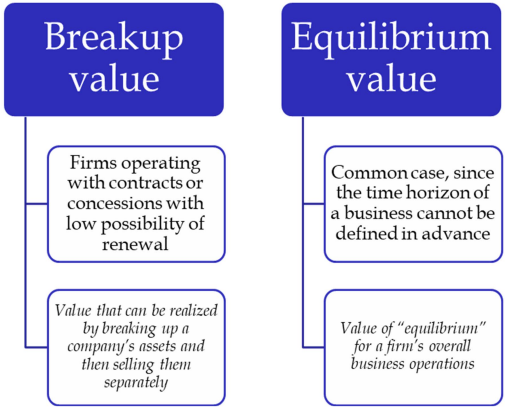
TV_n: Terminal Value after explicit projections of FCF

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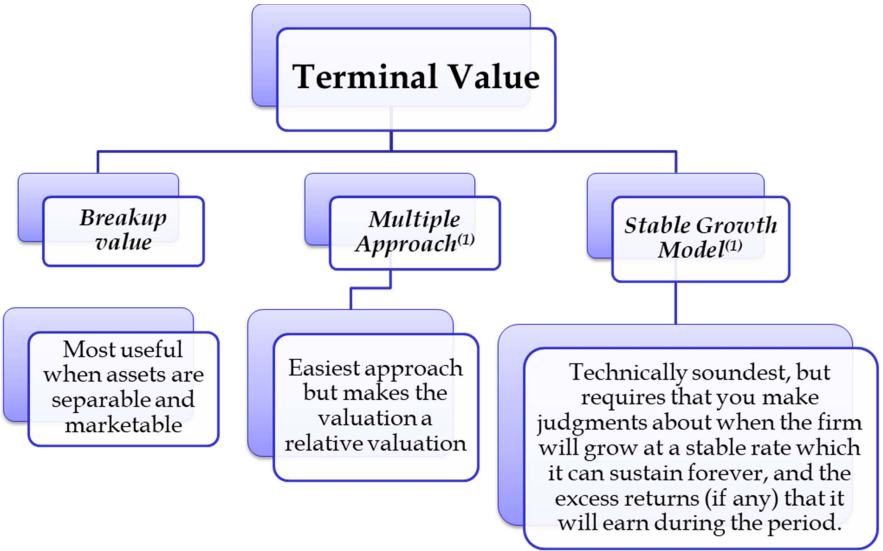
Terminal Value: Cash Flows Valuations II

A company's TV can assume two different meanings

1. The company breakup value (or liquidation value)
2. The equilibrium value of the company



Terminal Value: Estimation Approaches



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Terminal Value: Liquidation Value Approach

- In some valuations, we assume the firm will cease operations and sell its assets to *highest bidder*
- The resulting estimated assets value is the **Liquidation Value**

1. **Book Value (BV) Approach**
   - The Liquidation Value is the BV of assets at the end of the estimation period, adjusted for inflation (π) over the average age of the assets (s)

     $$\text{Expected Liquidation Value}_t = \text{BV of Assets}_t \cdot (1+\pi)^s$$

   - Subtract estimated debt value at end of the estimation period to get estimated equity value

     Limitation: based on accounting BV and not on assets' earnings power

2. **Earnings' Power Approach**
   - The Liquidation Value is the present value at appropriate discount rate r of expected CF from the assets over their remaining useful lives (j), after the explicit estimation period

     $$\text{Expected Liquidation Value}_t = \text{CF} \cdot a\neg_{j,r}$$

   - Subtract estimated debt value at end of estimation period to get estimated equity value

     Limitation: backed by a growth model with growth equal to zero

Terminal Value: Multiples Approach

- Firm value in a future year is estimated by applying a multiple to that firm's earnings or revenues in that year
  - For instance, in an asset side valuation

$$\text{TV}_n = \text{Revenue}_t \cdot \left(\frac{\text{EV}}{\text{Sales}} \right)_t$$

  - Subtract estimated debt value at end of estimation period to get estimated equity value, or
  - Use equity multiples (e.g., P/E) to arrive at the terminal (equity) value

- Although this has the virtue of simplicity, obtaining appropriate multiple is crucial
  - If multiple is estimated by looking at comparables, it becomes a relative valuation, not DCF
  - If multiple is estimated using fundamentals, it converges to the stable growth model (next)

- Using multiple estimated from pricing of comparables to determine TV can result in a dangerous mix of relative and DCF valuation
  - While there are advantages to relative valuation, a DCF valuation should provide you with an estimate of intrinsic value, not relative value

$\Longrightarrow$ The only consistent way of estimating TV in a DCF valuation is to use either: (i) a liquidation value or (ii) a stable growth model

Terminal Value: Stable Growth Model

- Unlike Liquidation Value, assuming firm has a finite life, Stable Growth model assumes firms reinvest CF into new assets (Firm Value = Assets in Place + Future Assets)

1. Two-stage Stable Growth Model

Assumption: FCF, beyond terminal year n, will grow at a constant rate g forever (i.e., Gordon Growth Model from $n+1$)

$$V_t = \sum_{s=t+1}^{n} \frac{\mathbb{E}_t\left[\text{FCF}_s\right]}{(1+k_C)^{s-t}} + \frac{\text{TV}_n}{(1+k_C)^n} = \sum_{s=t+1}^{n} \frac{\mathbb{E}_t\left[\text{FCF}_s\right]}{(1+k_C)^{s-t}} + \frac{\mathbb{E}_t\left[\text{FCF}_{n+1}\right]}{(k_C-g)\cdot(1+k_C)^n}$$

Equity Side Valuation: $\text{TV}_n = \frac{\mathbb{E}_t[\text{FCFE}_{n+1}]}{(\text{Cost of Equity}_{n+1}-g)}$, or $\frac{\mathbb{E}_t[\text{Dividends}_{n+1}]}{(\text{Cost of Equity}_{n+1}-g)}$

Asset Side Valuation: $\text{TV}_n = \frac{\mathbb{E}_t[\text{FCFO}_{n+1}]}{(\text{Cost of Capital}_{n+1}-g)}$

2. Three-stage Stable Growth Model

After the explicit estimation period, the firm will have (i) a period of high growth, (ii) a transition period of converging growth, (iii) a final period of constant growth forever

3. N-stage Growth Model

After the explicit estimation period, the firm's growth rate changes every year until final period

Stable Growth Rate

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Stable Growth Rate: Three Key Assumptions

1. **Length of the high growth period**
   a) Longevity: it is a matter of when, not whether: either firm becomes stable/large or is liquidated
   b) Firm size: smaller firms are more likely to earn excess returns than similar larger firms
   c) Ability to gain excess returns: growth depends on the ability to invest in new projects whose marginal return is higher than the cost of capital invested in such projects
   d) Competitive advantages: the bigger the competitive advantage and the stronger its sustainability, the more the firm is likely to produce excess returns (e.g., barriers to entry, management quality)

2. **The characteristics of stable growth firms**
   a) Equity risk: stable growth firms have lower β than higher growth firms (e.g. Fama-French model)
   b) Project returns: excess returns decline when reaching stable growth
   c) Debt ratios: stable growth firms tend to use more debt because as firms mature, debt capacity increases ($\rightarrow$ cost of capital)

3. **The transition to stable growth**
   a) Two-stage model: abrupt transition from high growth period to stable growth
   b) Three-stage model: intermediate transition period when its characteristics change gradually toward stable growth levels
   c) N-stage model: firm characteristics change each year from initial period to stable growth

Stable Growth Rate: Key Assumptions and Scenario

Which scenario to choose depends on the firm being valued

a) Two-stage model: best for firms with moderate growth rates, where the shift will not be too dramatic

b) Three-stage model: best for firms with very high growth rates in operating income

c) N-stage model: best for very young firms or firms with negative operating margins

Stable Growth Rate: Qualitative Aspects — I

- The emphasis on quantitative elements (e.g., ROC, margins, reinvestment rates, etc.) may strike some as skewed
- After all, growth is arguably also determined by a number of qualitative factors
  - The quality of management
  - The strength of a firm's marketing
  - Its capacity to form partnerships with other firms
  - The management's strategic vision
  - ...
- One might ask, where in the equations is there room for these factors?
  - Answer: qualitative factors do matter, but they all ultimately have to show up in one or more of the quantitative inputs that determine growth
    - Better managerial team should generate larger cash flows
    - More effective marketing team should drive higher sales
    - And so on

Stable Growth Rate: Qualitative Aspects — II

1. **Quality of management** affects:
   - (i) The ROC firms earn on new investments
   - (ii) How long they can sustain these returns
   - $\Longrightarrow$ Management quality may be one reason to justify why a firm's ROC can stay above CoC
2. **Marketing strength** affects:
   - (i) Operating margins
   - (ii) Turnover ratios
   - $\Longrightarrow$ Typically $\exists$ trade off lower margins vs higher turnover ratios; consider its implications for value
3. **Partnership with other firms** (acquisitions, R&D, etc.) affects:
   - Projected ROC: more effective firms/partnership strategies command and sustain higher ROC
4. **Strategic vision**:
   - Strategic vision shapes and is shaped by the strength of the competition that firms face
   - It determines size of excess returns (ROC $-$ CoC); and how quickly they fade to zero

- This way, qualitative factors are quantified into growth/valuation implications
  - Otherwise, you should remain skeptical about whether these factors truly affect value

Stable Growth Rate: Qualitative Aspects — III

- Why is it necessary to impose this quantitative structure on growth estimates?
  - One of the biggest dangers in valuing tech firms is using **story-telling** to justify growth rates that are neither: (i) *reasonable* nor (ii) *sustainable*

- For instance, you might be told:
  1. Tesla Motors will grow 100% a year because the "green" movement is strong
  2. Coca-Cola will grow 20% a year because it has a great brand name

- While there is a kernel of truth in these qualitative stories:
  - To what extent do these qualitative stories translate into a quantitative valuation model?
  - Do "green" tech or brand name support **higher future growth** than competitors?
    - The fact that an equity analyst knows firms/sectors they cover better than other analysts $\Longrightarrow$ growth estimates of these firms/sectors can be more precise than others' estimates
      $\not\Longrightarrow$ returns to investing in those firms/sectors will necessarily be better!

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Stable Growth Rate: Main Constraints

- The stable growth rate cannot exceed the economy growth rate; but it can be lower
  - It can even be negative; if so, the terminal value will be lower and you are assuming that your firm will "disappear" over time
  - In the long term, the real risk-free rate will converge to the real growth rate of the economy, and the nominal risk-free rate should approach the nominal growth rate of the economy

- The relation between the riskless rate, R_f, and the expected inflation rate, $\mathbb{E}[\pi]$, is

$$\text{Nominal } R_f = \text{Real } R_f + \mathbb{E}[\pi]$$

  - Simple rule of thumb: the stable growth rate should not exceed the riskless rate used

- Lastly, keep in mind:
  - Nominal vs. Real valuation consistency: nominal CF and discount rates should go with nominal growth rate; and vice versa
  - Currency consistency: the growth rate should be expressed in the currency in which the valuation is denominated

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Stable Growth Rate: Reinvestment Rate — I

- Linking reinvestment rate and retention ratio to the stable growth rate $\Longrightarrow$ valuation is **less sensitive** to assumptions about g
- To illustrate, increasing the stable growth rate, holding all else constant
  - Increases value directly, but changing the reinvestment rate as the growth rate changes will create an offsetting effect by reducing cash flows
  - The gains from increasing the growth rate will be partially or completely offset by the loss in cash flows because of the higher reinvestment rate
- Whether value increases or decreases as stable growth increases depends on assumptions on excess returns during stable growth period
  - ROC $> k_C \Longrightarrow$ increasing the stable growth rate will increase value
  - ROC $= k_C \Longrightarrow$ increasing the stable growth rate will have no effect on value

Stable Growth Rate: Reinvestment Rate — II

Terminal value for a firm with expected after-tax operating income of $100 million in year n+1 and a cost of capital of 10%

		ROC in perpetuity				
		6.00%	8%	10%	12.00%	14.00%
Growth Rate forever	0.0%	1000.00	1000.00	1000.00	1000.00	1000.00
	0.5%	965	987	1000	1009	1015
	1.0%	926	972	1000	1019	1032
	1.5%	882	956	1000	1029	1050
	2.0%	833	938	1000	1042	1071
	2.5%	778	917	1000	1056	1095
	3.0%	714	893	1000	1071	1122

Stable Growth Rate: Reinvestment Rate — III

- When a firm reaches stable growth, returns in excess of cost of capital decline. Eventually, if excess returns go to zero, the stable growth rate does not affect valuation:

$$\begin{aligned} \text{TV}_n = \frac{\mathbb{E}_t\left[\text{FCF}_{n+1}\right]}{k_C - g} &= \frac{\mathbb{E}_t\left[\text{EBIT}_{n+1} \cdot (1-\tau_C) - \text{Net CAPEX}_{n+1} - \Delta\text{WC}_{n+1}\right]}{k_C - g} \\ &= \frac{\mathbb{E}_t\left[\text{EBIT}_{n+1} \cdot (1-\tau_C) \cdot (1 - \text{Reinvestment Rate})\right]}{k_C - g} \\ &= \frac{\mathbb{E}_t\left[\text{EBIT}_{n+1} \cdot (1-\tau_C) \cdot (1 - \text{Reinvestment Rate})\right]}{k_C - \text{Reinvestment Rate} \cdot \text{ROC}} \end{aligned}$$

- If Cost of Capital k_C = ROC (and dropping $\mathbb{E}_t\left[.\right]$ for ease of illustration)

$$\begin{aligned} \text{TV}_n &= \frac{\text{EBIT}_{n+1} \cdot (1-\tau_C) \cdot (1 - \text{Reinvestment Rate})}{k_C - \text{Reinvestment Rate} \cdot k_C} \\ &= \frac{\text{EBIT}_{n+1} \cdot (1-\tau_C) \cdot (1 - \text{Reinvestment Rate})}{k_C \cdot (1 - \text{Reinvestment Rate})} = \frac{\text{EBIT}_{n+1} \cdot (1-\tau_C)}{k_C} \end{aligned}$$

- Similar arguments (and derivations) apply to DDM and FCFE

Takeaways

- Assumptions for the constant growth perpetuity model include
  - Positive cash flow next period
  - Constant discount rate – implied constant capital structure ratios
  - Constant growth rate less than discount rate
- What is the reasonable range of growth rates to use in perpetuity?
  - It can be positive but has to be sustainable
    - Long-run inflation
    - Growth in the economy (inflation and real)
    - Productivity gains
  - It can be negative
    - Decreasing margins due to price pressure, increased costs, or loss of competitive advantage
- Assess reasonableness of growth
  - Use basic financial analysis tools to assess underlying assumptions
  - Analyze growth rates of comparables from financial analysts or FCF forecasts
  - Analyze implied market multiples; compare them to current multiples of comparables
  - Consider how current market multiples will evolve to the continuing value date